

$\therefore$ The condition that for each row the leading entry must be in a column to the right of the columns of the leading entries of all rows above it is True.

3. Row 2 Column 1 = Row 3 Column 1 = 0

Row 3 Column 2 = 0

$\therefore$ The condition that for each column having a leading entry, all positions below it must contain 0 is True.

4. The condition that each leading entry must be 1 is True.

5. Row 1 Column 3 = Row 2 Column 3 = 0

Row 1 Column 2 = 0

$\therefore$ The condition that for each column having a leading entry, all positions above it must contain 0 is True.

$\therefore$ The matrix $\begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix}$ is in rref $\square$

$\therefore$ By defn, each of the columns of A are pivot columns.

$\therefore$ By defn, there are no free variables.

From Theorem 2, $\therefore$ there are no rows in the matrix of the form $[0 \ 0 \ 0 \ \dots \ 0 \ b]$, where $b \neq 0$. $\therefore$ The system is consistent and has a unique solution. $\square$

The corresponding system of equations are: $x_1 = 3, x_2 = 1, x_3 = 2$. $\therefore$ The vector x is $\begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$.

5. $A = \begin{bmatrix} 1 & -5 & -7 \\ -3 & 7 & 5 \end{bmatrix}, b = \begin{bmatrix} -2 \\ -2 \end{bmatrix}$

Solution: We need to find the vector x s.t. $T(x) = b$. $\therefore Ax = b$

By Theorem 3, the matrix equation $Ax = b$ has the same solution set as the system of equations corresponding to the augmented matrix $[A \ b]$.

$$\begin{bmatrix} 1 & -5 & -7 & -2 \\ -3 & 7 & 5 & -2 \end{bmatrix} \xrightarrow[\substack{+3 \times \text{Row 1}}]{\text{Row 2} = \text{Row 2}} \begin{bmatrix} 1 & -5 & -7 & -2 \\ 0 & -8 & -16 & -8 \end{bmatrix} \xrightarrow[\substack{\times (-1/8)}]{\text{Row 2} = \text{Row 2}} \begin{bmatrix} 1 & -5 & -7 & -2 \\ 0 & 1 & 2 & 1 \end{bmatrix}$$

$$\xrightarrow[\substack{+5 \times \text{Row 2}}]{\text{Row 1} = \text{Row 1}} \begin{bmatrix} 1 & 0 & 3 & 3 \\ 0 & 1 & 2 & 1 \end{bmatrix}$$